Automated Contrast Correction System for Under/Over-Exposed Photography Using Optimized Histogram Equalization

Course: COMP7116001 — Computer Vision
Semester: Even 2025/2026
Lecturer: Fiqri Ramadhan Tambunan, S.Kom., M.Kom

Team 2: | NIM | Name | Role | |—|—|—| | 2802486980 | Ahmad Daffa Hidayatullah | | | 2802493752 | Andrey Apriliady | | | 2802481626 | Ezra Mayurga | | | 280248447 | Jason Firenze Trianto | | | 2802488185 | Keanu Stadeva | | | 2802484483 | Rafifdiya | |

Abstract

Poor exposure in digital photography — whether under-exposure in low-light environments or over-exposure in bright conditions — significantly degrades image quality and reduces the utility of captured images. This paper presents an automated contrast correction system that applies optimized histogram equalization techniques to enhance image quality across both exposure types. The system employs Global Histogram Equalization (GHE) as a baseline and Contrast Limited Adaptive Histogram Equalization (CLAHE) as the primary enhancement method, both operating exclusively on the Lightness (L) channel of the CIELAB color space to preserve color fidelity. Enhancement is performed through a BGR→LAB→equalize L→LAB→BGR pipeline. Additionally, two machine learning modules — an Exposure Classifier and a Parameter Predictor — are trained using Random Forest on 10 histogram-based features extracted from the L channel, enabling automatic detection of exposure condition and prediction of optimal CLAHE parameters. The system is deployed as an interactive web application using Streamlit. Experiments on the LOL (Low-Light) dataset (485 training pairs, 15 test pairs) show that the optimal CLAHE configuration (clip limit = 4.0, tile size = 4×4) achieves a mean SSIM of 0.4519 and PSNR of 10.54 dB, outperforming GHE in structural similarity (SSIM: 0.4431). The exposure classifier achieves 86.74% cross-validation accuracy, and the parameter predictor achieves a clip limit MAE of 0.0837 and tile size classification accuracy of 99.59%.

---

1. Introduction

1.1 Background

Digital photography has become ubiquitous in modern life, with billions of images captured daily via smartphones, surveillance cameras, medical imaging devices, and autonomous vehicles. However, real-world imaging conditions often deviate from ideal exposure settings. Scenes with insufficient lighting — such as night photography, indoor environments, or backlit subjects — produce under-exposed images where dark regions lose critical detail. Conversely, scenes with intense illumination — such as direct sunlight or flash photography — produce over-exposed images where bright highlights are "blown out" and structural information is lost.

Contrast enhancement through histogram manipulation is among the most established approaches to correcting these exposure deficiencies. Global Histogram Equalization (GHE) and Contrast Limited Adaptive Histogram Equalization (CLAHE) are two fundamental techniques widely used in this domain. However, their performance is highly sensitive to parameter selection, and neither method automatically adapts to the specific exposure characteristics of an input image.

1.2 Problem Statement

Existing histogram equalization methods present three key challenges:

1. **Parameter sensitivity:** CLAHE performance varies significantly with clip limit and tile grid size. Suboptimal parameters produce artifacts or insufficient enhancement.
2. **No exposure awareness:** Standard implementations do not distinguish between under-exposed and over-exposed images, applying the same processing indiscriminately.
3. **Manual tuning burden:** Users without domain knowledge cannot effectively select appropriate parameters for their images.

1.3 Objectives

This project aims to:

1. Implement and compare GHE and CLAHE on the LAB color space for exposure-preserving contrast correction.
2. Empirically determine optimal CLAHE parameters through systematic grid search on the LOL dataset.
3. Train a machine learning classifier to automatically identify image exposure condition.
4. Train a machine learning predictor to automatically select optimal CLAHE parameters per image.
5. Deploy the complete system as an accessible web application.

1.4 Significance

Automated exposure correction has broad practical applications: improving visibility in surveillance footage, pre-processing medical images for diagnosis, enhancing photographs for social media, and preparing input images for downstream computer vision pipelines (object detection, face recognition). By integrating machine learning into the parameter selection pipeline, this system reduces the need for manual tuning while maintaining enhancement quality.

---

2. Related Work

2.1 Histogram Equalization

Histogram equalization redistributes pixel intensity values to achieve a more uniform distribution, thereby increasing global contrast. Pizer et al. (1987) introduced CLAHE as an improvement over GHE, addressing GHE's tendency to over-amplify noise in homogeneous regions by limiting contrast amplification through a clip limit parameter and operating on local image tiles rather than the global histogram [1].

2.2 Color Space Considerations

Direct histogram equalization on RGB channels alters hue and saturation, producing unnatural color shifts. Applying equalization exclusively to the luminance channel of a perceptually uniform color space (such as CIELAB or HSV) decouples brightness enhancement from color information. Huang et al. demonstrated that L-channel equalization in LAB space produces more visually natural results than RGB-based methods [2].

2.3 Low-Light Image Enhancement

The Low-Light (LOL) dataset introduced by Wei et al. (2018) provides paired low-light and normal-light images for benchmarking enhancement algorithms. State-of-the-art deep learning methods such as RetinexNet and EnlightenGAN achieve SSIM values of 0.55–0.75 on this dataset [3]. Classical histogram-based methods typically achieve SSIM of 0.40–0.50, consistent with our experimental findings.

2.4 Machine Learning for Image Quality

Random Forest classifiers have been successfully applied to image quality assessment tasks using handcrafted features derived from statistical properties of pixel intensity distributions [4]. Features such as mean brightness, standard deviation, skewness, kurtosis, and percentile values of the luminance histogram provide discriminative power for exposure classification without the computational overhead of deep neural networks.

---

3. Methodology

3.1 Dataset

The LOL (Low-Light) dataset [3] is used for all experiments. It contains paired images of identical scenes captured under low-light and normal-light conditions:

Split	Images
Training	485 pairs (low + normal)
Test	15 pairs (low + normal)

Images are PNG format, varying resolutions. The dataset is publicly available under academic use license. No personal or sensitive data is included.

3.2 Core CV Pipeline

All image processing follows a LAB-channel pipeline to preserve color fidelity:

```
Input (BGR) → BGR2LAB → Split L, A, B → Equalize L → Merge → LAB2BGR → Output (BGR)
```

By operating exclusively on the L (Lightness) channel, hue and saturation encoded in the A and B channels remain unchanged throughout processing.

3.2.1 Global Histogram Equalization (GHE) GHE applies OpenCV's `equalizeHist()` to the L channel. The transformation function is the cumulative distribution function (CDF) of the L channel histogram:

```
s = T(r) = (L_max / MN) * Σ n_j   for j = 0 to r
```

where M×N is the image resolution and n_j is the pixel count at intensity j. GHE serves as the baseline for comparison.

3.2.2 CLAHE CLAHE (Contrast Limited Adaptive Histogram Equalization) divides the L channel into non-overlapping tiles and applies histogram equalization locally within each tile. The clip limit parameter bounds the histogram height before redistribution, preventing over-amplification of noise. Bilinear interpolation at tile boundaries eliminates blocking artifacts.

Key parameters: - **Clip Limit:** Controls contrast amplification cap (range: 1.0–8.0) - **Tile Size:** Grid dimensions for local equalization (options: 4×4, 8×8, 16×16)

3.3 Evaluation Metrics

3.3.1 Peak Signal-to-Noise Ratio (PSNR) PSNR measures pixel-level fidelity between enhanced output and ground truth:

```
PSNR = 10 * log10(MAX² / MSE)
```

Higher PSNR indicates greater pixel-level similarity. Typical range: >20 dB considered acceptable.

3.3.2 Structural Similarity Index (SSIM) SSIM evaluates perceptual similarity by comparing luminance, contrast, and structural patterns between images:

```
SSIM(x,y) = [l(x,y)]^ * [c(x,y)]^ * [s(x,y)]^
```

Range: 0–1, where 1 indicates perfect similarity. SSIM is considered more aligned with human visual perception than PSNR.

3.4 Grid Search Experiment

A systematic grid search evaluates all 12 CLAHE parameter combinations across the 485 training images:

Parameter	Values Tested
Clip Limit	1.0, 2.0, 3.0, 4.0
Tile Size	4×4, 8×8, 16×16

For each image, the combination yielding the highest SSIM is recorded as the optimal parameter pair.

3.5 Machine Learning Modules

3.5.1 Feature Extraction Ten features are extracted from the L-channel histogram of each input image:

Feature	Description
Mean	Average L-channel intensity
Std	Standard deviation of intensity
Skewness	Distribution asymmetry
Kurtosis	Distribution peakedness
P10, P25, P75, P90	Percentile values
Contrast Ratio	P90 − P10
Entropy	Information content of histogram

3.5.2 Auto Exposure Classifier A Random Forest Classifier (100 trees) is trained on three exposure classes:

- **Underexposed:** 485 samples from LOL `train/low/`
- **Normal:** 485 samples from LOL `train/normal/`
- **Overexposed:** 485 synthesized samples (gamma=0.5 applied to normal images)

Overexposed samples are synthesized via gamma correction (=0.5) applied to normal images via a lookup table transformation, simulating highlight clipping and brightness inflation characteristic of over-exposure.

A hybrid rule-based post-processing step augments the classifier: images predicted as "normal" with p90 > 240 and mean > 128 are reclassified as "overexposed", capturing blown-highlight cases where the synthesized training distribution underrepresents real-world overexposure.

3.5.3 Auto Parameter Predictor From the grid search results, each training image is labeled with its optimal CLAHE parameters (highest SSIM). Two models are trained:

- **Clip Limit Regressor:** Random Forest Regressor predicts clip limit value
- **Tile Size Classifier:** Random Forest Classifier predicts tile size (4, 8, or 16)

At inference, the predicted clip limit is rounded to the nearest 0.5 and clamped to [1.0, 8.0].

---

4. Implementation & Results

4.1 System Architecture

The system is implemented in Python using the following stack:

Component	Library/Tool
Image processing	OpenCV 4.9.0
Evaluation metrics	scikit-image 0.23.2
Machine learning	scikit-learn 1.4.2
Statistical features	scipy 1.13.0
Web application	Streamlit 1.35.0
Model persistence	joblib 1.4.2

4.2 Grid Search Results

Table 1 presents mean PSNR and SSIM averaged over 485 training images for each parameter combination and GHE baseline.

Table 1: CLAHE Grid Search Results (Mean over 485 train images)

Method	Clip Limit	Tile Size	PSNR (dB)	SSIM
GHE (baseline)	—	—	**15.33**	0.4431
CLAHE	1.0	4×4	8.61	0.2933

Method	Clip Limit	Tile Size	PSNR (dB)	SSIM
CLAHE	1.0	8×8	8.50	0.2836
CLAHE	1.0	16×16	8.38	0.2705
CLAHE	2.0	4×4	9.27	0.3659
CLAHE	2.0	8×8	9.07	0.3506
CLAHE	2.0	16×16	8.87	0.3328
CLAHE	3.0	4×4	9.91	0.4171
CLAHE	3.0	8×8	9.56	0.3954
CLAHE	3.0	16×16	9.30	0.3768
CLAHE	**4.0**	**4×4**	10.54	**0.4519**
CLAHE	4.0	8×8	10.07	0.4275
CLAHE	4.0	16×16	9.71	0.4059

Key finding: CLAHE with clip=4.0, tile=4×4 achieves the highest SSIM (0.4519), surpassing GHE (0.4431). GHE achieves higher PSNR (15.33 dB) due to its global histogram mapping producing closer pixel-level values, but SSIM — which better reflects human perceptual quality — favors CLAHE.

The trend shows: higher clip limits consistently improve both PSNR and SSIM; smaller tile sizes (4×4) outperform larger grids, suggesting finer local adaptation benefits the LOL dataset's low-light images.

4.3 Machine Learning Results

Table 2: ML Model Performance

Model	Metric	Value
Exposure Classifier	CV Accuracy (5-fold)	86.74% ± 4.65%
Clip Limit Predictor	MAE (5-fold CV)	0.0837
Tile Size Classifier	CV Accuracy (5-fold)	99.59%

The exposure classifier exceeds the 85% accuracy target. The exceptionally high tile size accuracy (99.59%) indicates that the optimal tile size is highly predictable from histogram features — most images in the LOL dataset consistently favor smaller tiles.

Validation on 8 real-world photos (3 underexposed, 3 overexposed, 2 normal) confirms correct classification for all 8 samples, demonstrating generalization beyond the LOL training distribution.

4.4 Web Application

The Streamlit application provides: - **Input:** File upload or live camera capture - **Processing:** Parallel GHE and CLAHE enhancement - **ML Integration:**

Exposure badge (underexposed/normal/overexposed) + auto parameter selection - **Evaluation:** PSNR and SSIM metrics when ground truth is provided - **Visualization:** Side-by-side image comparison + L-channel histogram plots - **Output:** Downloadable PNG results (GHE and CLAHE) - **Mode toggle:** Manual (slider control) vs. Auto ML (model-driven parameters)

---

5. Discussion & Limitations

5.1 PSNR vs. SSIM Trade-off

A notable observation is the inverse relationship between PSNR and SSIM across methods. GHE achieves superior PSNR (15.33 dB) while CLAHE achieves superior SSIM (0.4519). This reflects a fundamental trade-off: GHE's global equalization produces pixel values statistically closer to ground truth (higher PSNR), but CLAHE's local adaptation better preserves structural patterns (higher SSIM). For perceptual quality assessment, SSIM is the more relevant metric, supporting CLAHE as the preferred method.

5.2 Performance vs. Deep Learning

All SSIM values (0.27–0.45) are below those of state-of-the-art deep learning methods (0.55–0.75 on LOL). This is expected — histogram-based classical methods lack the representational capacity of neural networks. The advantage of the proposed system is computational efficiency: inference requires no GPU and runs in milliseconds, making it suitable for real-time applications and resource-constrained environments.

5.3 Overexposed Class Imbalance

The training set for the exposure classifier lacks real overexposed samples; the overexposed class is entirely synthesized via gamma correction. While effective on test samples (8/8 correct), the distribution of synthesized overexposed images may not fully represent real-world overexposure artifacts (specular reflections, bloom effects, JPEG compression artifacts in highlights). This is mitigated by the hybrid rule-based fallback.

5.4 Dataset Scope

The LOL dataset consists exclusively of low-light images. The "normal" class represents correctly-exposed indoor/outdoor scenes that may not cover the full spectrum of real-world photography conditions (HDR scenes, flash photography, medical imaging). Performance may vary for images significantly outside the training distribution.

5.5 Single-Channel Limitation

Operating on the L channel only prevents correction of chromatic aberrations or color casts introduced by non-neutral lighting. Images with strong color temperature shifts (e.g., incandescent lighting) may show improved luminance but residual color imbalance.

---

6. Conclusion & Future Work

6.1 Conclusion

This paper presented an automated contrast correction system combining classical histogram equalization (GHE, CLAHE) with machine learning for exposure classification and parameter prediction. The system achieves:

- CLAHE with clip=4.0, tile=4×4 as the empirically optimal configuration (SSIM=0.4519, PSNR=10.54 dB on 485 training images)
- CLAHE outperforms GHE in SSIM (+2.0%), confirming superior perceptual quality
- Exposure classifier accuracy of 86.74%, exceeding the 85% target
- Parameter predictor MAE of 0.0837 (clip limit) and 99.59% accuracy (tile size)
- 8/8 correct classifications on real-world photos (underexposed, overexposed, normal)

The web application integrates all components into an accessible interface that removes the need for manual parameter tuning.

6.2 Future Work

1. **Deep learning integration:** Incorporate lightweight neural networks (e.g., Zero-DCE, MIRNet) as an optional enhancement mode for higher SSIM.
2. **Real overexposed training data:** Augment the classifier with real overexposed images rather than relying solely on gamma synthesis.
3. **Multi-scale CLAHE:** Explore adaptive tile size selection based on image resolution.
4. **Video support:** Extend the pipeline to process video streams with temporal consistency constraints.
5. **Mobile deployment:** Convert the model to TFLite or ONNX for on-device mobile inference.

References

[1] S. M. Pizer et al., "Adaptive Histogram Equalization and Its Variations,"

Computer Vision, Graphics, and Image Processing, vol. 39, no. 3, pp. 355–368, 1987.

[2] S.-C. Huang, F.-C. Cheng, and Y.-S. Chiu, "Efficient Contrast Enhancement Using Adaptive Gamma Correction With Weighting Distribution," *IEEE Transactions on Image Processing*, vol. 22, no. 3, pp. 1032–1041, 2013.

[3] C. Wei, W. Wang, W. Yang, and J. Liu, "Deep Retinex Decomposition for Low-Light Enhancement," *British Machine Vision Conference (BMVC)*, 2018. Dataset: https://github.com/weichen582/RetinexNet

[4] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[5] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image Quality Assessment: From Error Visibility to Structural Similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.

Appendix

A. Team Contribution Statement

NIM	Name	Contributions
2802486980	Ahmad Daffa Hidayatullah	*(fill in)*
2802493752	Andrey Apriliady	*(fill in)*
2802481626	Ezra Mayurga	*(fill in)*
280248447	Jason Firenze Trianto	*(fill in)*
2802488185	Keanu Stadeva	*(fill in)*
2802484483	Rafifdiya	*(fill in)*

B. Screenshots

(Insert screenshots of Streamlit application: upload interface, GHE/CLAHE comparison, histogram visualization, ML exposure badge, metrics display)

C. Code Snippets

Core pipeline (`processing/ghe.py`):

```
def apply_ghe(image: np.ndarray) -> np.ndarray:
    L, A, B = bgr_to_lab(image)
    L_eq = cv2.equalizeHist(L)
    return lab_to_bgr(L_eq, A, B)
```

CLAHE application (`processing/clahe.py`):

```python
def apply_clahe(image, clip_limit=2.0, tile_size=(8,8)):
    L, A, B = bgr_to_lab(image)
    clahe = cv2.createCLAHE(clipLimit=clip_limit, tileGridSize=tile_size)
    L_eq = clahe.apply(L)
    return lab_to_bgr(L_eq, A, B)
```

Feature extraction (`ml/features.py`):

```python
def extract_features(image: np.ndarray) -> np.ndarray:
    lab = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)
    L = cv2.split(lab)[0].ravel().astype(np.float32)
    mean, std = np.mean(L), np.std(L)
    skewness, kurt = stats.skew(L), stats.kurtosis(L)
    p10, p25, p75, p90 = np.percentile(L, [10, 25, 75, 90])
    contrast = p90 - p10
    hist, _ = np.histogram(L, bins=256, range=(0,256), density=True)
    entropy = -np.sum((hist+1e-10) * np.log2(hist+1e-10))
    return np.array([mean, std, skewness, kurt, p10, p25, p75, p90, contrast, entropy])
```

Report prepared by Team 2 — COMP7116001 Computer Vision — Binus University — 2025/2026